

OSM SOG36 (P)

HIGHLIGHTS

- **2x Arm Cortex-A76 + 6x Arm Cortex-A55** deliver high performance and power efficiency, built on TSMC 6nm process
- MediaTek 8th generation **NPU with 7.4 TOPS, total up to 8.5 TOPS** acceleration, enabling **real-time AI**, GenAI applications with **leading LLMs**
- **Arm Mali-G57 MC2 GPU** and advanced multimedia pipelines support up to **4K display and 4K video codecs**

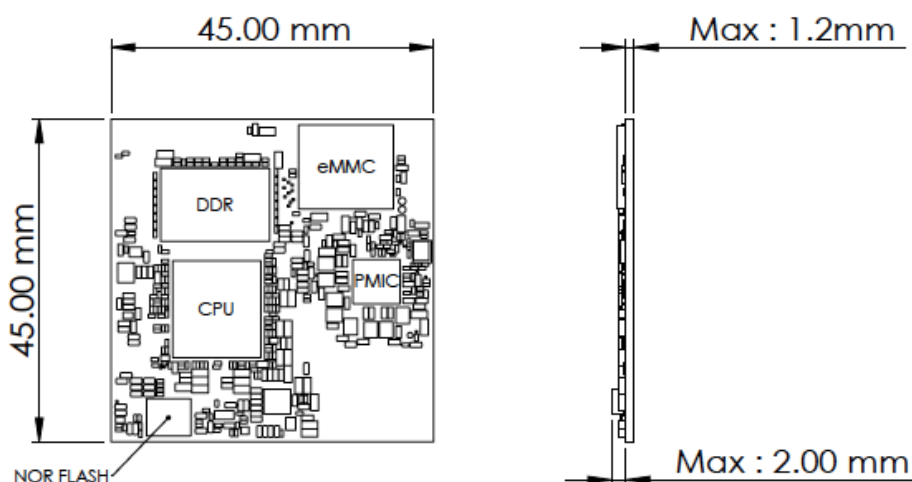
FEATURE

- Power-Efficient Edge GenAI Platform
- Advanced GenAI and Neural Processing
- Advanced Multimedia and Graphics
- Android 16 & Yocto 5.0 support

POTENTIAL APPLICATIONS

- Smart Home
- Industrial
- Smart Retail
- Enterprise
- Healthcare
- Smart Cities

BOARD PLACEMENT



SPECIFICATION

Processor

CPU (SOG36) 1x Arm Cortex-A76 2.0GHz
5x Arm Cortex-A55 2.0GHz

CPU (SOG36P) 2x Arm Cortex-A76 2.0GHz
6x Arm Cortex-A55 2.0GHz

GPU (SOG36) Arm Mali-G57 MC2

GPU (SOG36P) Arm Mali-G57 MC2

NPU (SOG36P) 8th-Gen NPU 7.4 TOPS

NPU (SOG36) 8th-Gen NPU 5.1 TOPS

Memory and Type

LPDDR4(X)-4GB / 8GB

Storage Type

eMMC5.1 64GB / 32GB

Display Interface

1x MIPI DSI 4-lane

1x eDP1.2 4-lane

1x DP1.4 4-lane

Image Signal Processor

1x ISP (3A, NR, AI-FD, LSC, Warp Engine)

Single camera: 16MP @ 30fps

Audio

2x I2SIN

2x I2SOUT

Camera

2x MIPI CSI 4-lane

Mechanical & Environmental

Form Factor 45 mm x 45 mm x 3.2 mm

Temp. Range -20°C to 80°C

Input/Output

2x USB 2.0 (host)

1x USB 3.2 Gen1(DRD, combo with DP1.4)

1x Ethernet

3x PWM

36x GPIOs

1x PCIe Gen2 1-lane

1x UART with HW flow control

3x UART(1x console)

10x I2C

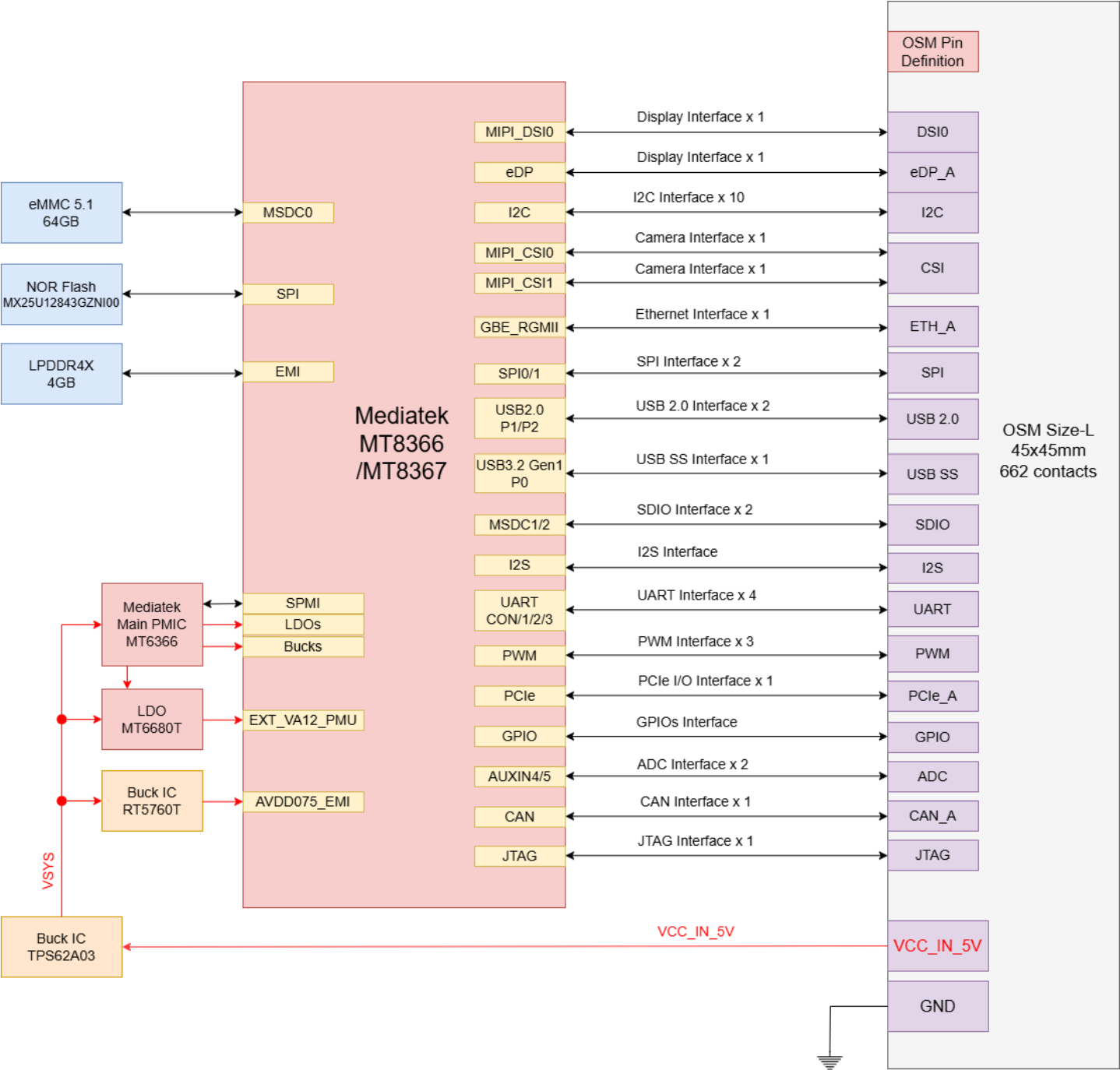
2x SPI

2x ADC

2x SDIO

*** Note: The difference between SOG36P and SOG36 is the number of CPU cores; all other specifications are identical.**

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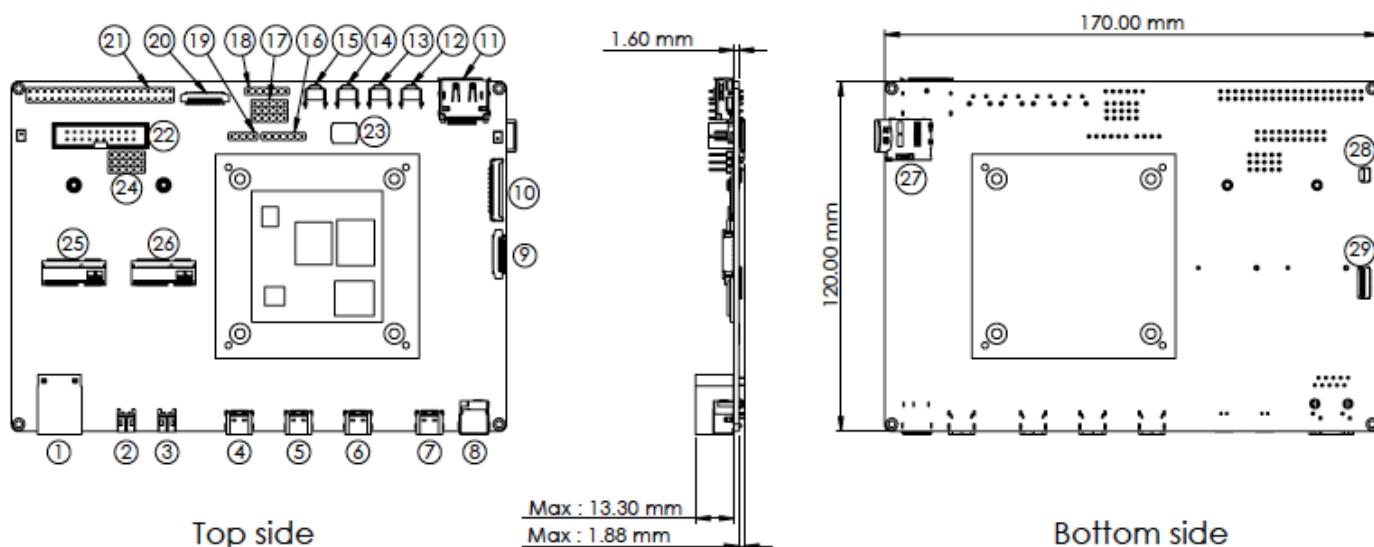


OSM Carrier Board

FEATURE

- Supports **one ultra-wide 4K display** or **dual 2.5K displays**.
- External support for **Wi-Fi 5 (MT-7663)** or **Wi-Fi 6 (MT-7921)** via SDIO or PCIe modules.
- Single camera 16MP @30fps

BOARD PLACEMENT



No.	Function	No.	Function	No.	Function
1	RJ45	11	Display Port	21	Raspberry Pi
2	SPK R	12	Power Key	22	JTAG
3	SPK L	13	Reset Key	23	Display SEL / Boot SEL / RSPI_EN
4	USB Type C (A)	14	Vol up / Download	24	PCM SEL
5	USB Type C (C)	15	Vol down	25	M.2 E key PCIe
6	UART Debug Port	16	SPI_A	26	M.2 E key SDIO_B
7	USB Type C (Power)	17	I2S_SEL	27	SD Card
8	DC jack	18	I2SIN	28	Touch panel interface
9	CAM_B	19	I2C	29	DSI interface
10	eDP	20	CAM_A		

OSM Carrier Board

SPECIFICATION

Compatibility	Open Standard Module v1.2
Network	1x RJ45 GbE
Main Display	1x MIPI DSI 4-lane 1x DP1.4 4-lane
Second Display	1x DP1.4 4-lane (Type C) 1x eDP1.2 4-lane
Camera	2x 4-lane MIPI-CSI
Audio	2x SPK connector (Audio AMP AD82129) 1x I2SIN 6PIN connector 2x DMIC
USB	1x USB 3.2 gen1 Type C (DRD & DPoC/DP Alt mode) 1x USB 2.0 Type C (host)
Debug Port	1x USB2.0 Type C interface connector
SPI	1x SPI 6PIN connector (1.8V)
I2C	1x I2C 4PIN connector (1.8V)
Storage	1x microSD slot
RSPi	1x 40PIN RSPi connector
JTAG	1x 20PIN JTAG connector (Shared by GPIOs of RSPi)
Switch & Buttons	1x Power Button 1x Reset Button 1x Vol Up/Download 1x Vol Down
Wireless	2x M.2 2230 key E slot
Power	DC JACK 12V-IN USB Type C PD
Dimension	170mm * 120mm
Operating Temperature	0° C to +70 ° C

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